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Applied Economics

ISSN: 0003-6846 (Print) 1466-4283 (Online) Journal homepage: www.tandfonline.com/journals/raec20

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To cite this article: Szabolcs Blazsek, Han-Chiang Ho & Su-Ping Liu (2018) Score-driven Markov-switching EGARCH models: an application to systematic risk analysis, Applied Economics, 50:56, 6047-6060, DOI: 10.1080/00036846.2018.1488073

To link to this article: https://doi.org/10.1080/00036846.2018.1488073

Published online: 29 Aug 2018.

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APPLIED ECONOMICS
2018, VOL. 50, NO. 56, 6047–6060
https://doi.org/10.1080/00036846.2018.1488073

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Score-driven Markov-switching EGARCH models: an application to systematic risk analysis

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ABSTRACT

We introduce new Markov-switching (MS) dynamic conditional score (DCS) exponential generalized autoregressive conditional heteroscedasticity (EGARCH) models, to be used by practitioners for forecasting value-at-risk (VaR) and expected shortfall (ES) in systematic risk analysis. We use daily log-return data from the Standard & Poor's 500 (S&P 500) index for the period 1950–2016. The analysis of the S&P 500 is useful, for example, for investors of (i) well-diversified US equity portfolios; (ii) S&P 500 futures and options traded at Chicago Mercantile Exchange Globex; (iii) exchange traded funds (ETFs) related to the S&P 500. The new MS DCS-EGARCH models are alternatives to of the recent MS Beta-*t*-EGARCH model that uses the symmetric Student's *t* distribution for the error term. For the new models, we use more flexible asymmetric probability distributions for the error term: Skew-Gen-*t* (skewed generalized *t*), EGB2 (exponential generalized beta of the second kind) and NIG (normal-inverse Gaussian) distributions. For all MS DCS-EGARCH models, we identify high- and low-volatility periods for the S&P 500. We find that the statistical performance of the new MS DCS-EGARCH models is superior to that of the MS Beta-*t*-EGARCH model. As a practical application, we perform systematic risk analysis by forecasting VaR and ES.

Abbreviation Single regime (SR); Markov-switching (MS); dynamic conditional score (DCS); exponential generalized autoregressive conditional heteroscedasticity (EGARCH); value-at-risk (VaR); expected shortfall (ES); Standard & Poor's 500 (S&P 500); exchange traded funds (ETFs); Skew-Gen-t (skewed generalized t); EGB2 (exponential generalized beta of the second kind); NIG (normal-inverse Gaussian); log-likelihood (LL); standard deviation (SD); partial autocorrelation function (PACF); likelihood-ratio (LR); ordinary least squares (OLS); heteroscedasticity and autocorrelation consistent (HAC); Akaike information criterion (AIC); Bayesian information criterion (BIC); Hannan-Quinn criterion (HQC).

KEYWORDS

Dynamic conditional score (DCS) models; Markov-switching (MS) models; value-at-risk (VaR); expected shortfall (ES); systematic risk analysis

JEL CLASSIFICATION

C22; C52; C58

I. Introduction

In this article, we suggest the use of new Markov-switching (MS) dynamic conditional score (DCS) exponential generalized autoregressive conditional heteroscedasticity (EGARCH) models, to be applied by practitioners for forecasting value-at-risk (VaR) (Jorion 2006) and expected shortfall (ES) (Acharya et al., 2012, 2017) in systematic risk analysis. We use daily log-return data from the Standard & Poor's 500 (S&P 500) for the period 1950–2016. The analysis of the S&P 500 is useful, for example, for investors of (i) well-diversified US equity portfolios (Bahr and Maas 2014); (ii) S&P 500 futures and options traded at Chicago Mercantile Exchange Globex (Schwartz, Van Ness, and Van Ness 2004); (iii) exchange traded funds (ETFs) related to the S&P 500 (Xu and Yin 2017). The new MS DCS-EGARCH models of this article are alternatives to the recent MS Beta-*t*-EGARCH model (Blazsek and Ho 2017) that uses the symmetric Student's *t* distribution with a regime-switching degrees of freedom parameter for the error term. Blazsek and Ho (2017) extend the single-regime Beta-*t*-EGARCH model of Harvey and Chakravarty (2008). By using daily S&P 500 log-return data, Blazsek and Ho (2017) show that the likelihood-based statistical performance of the MS Beta-*t*-EGARCH model is superior to that of the single-regime Beta-*t*-EGARCH model.

For the new MS DCS-EGARCH models, we use flexible asymmetric probability distributions for the error term: Skew-Gen-*t* (skewed generalized *t*), EGB2 (exponential generalized beta of the second kind) and NIG (normal-inverse Gaussian) (Barndorff-Nielsen and Halgreen 1977) distributions (to the best of our knowledge, the use of the NIG distribution is new in

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the body of DCS literature). Thus, in this article, we introduce the MS Beta-Skew-Gen-t-EGARCH, MS EGB2-EGARCH and MS NIG-EGARCH models, respectively. Single-regime versions of Beta-Skew-Gen-t-EGARCH and EGB2-EGARCH are suggested in the works of Harvey and Lange (2017) and Caivano and Harvey (2014), respectively.

We compare the new MS DCS-EGARCH models with the MS Beta-t-EGARCH model, by considering leverage effects (Black 1976; Harvey 2013) in all DCS-EGARCH models. For all MS models, we use two switching regimes and a constant transition probability matrix. It is straightforward to extend the mathematical formulas of model specification and statistical inference to multi-regime MS DCS-EGARCH. We consider unrestricted and restricted specifications of all MS models. For the unrestricted MS models, all parameters may take different values in different regimes. For the restricted MS models, some parameters take the same values in both regimes. We define the restricted specifications, by using robust statistical tests of parameter differences. For all MS DCS-EGARCH models, we identify high- and low-volatility periods for the S&P 500. We find that the statistical performance of each MS DCS-EGARCH model is superior to that of the single-regime alternative. We also find that the new MS DCS-EGARCH models are superior to the MS Beta-t-EGARCH model. Finally, as a practical application, we perform a systematic risk analysis by forecasting VaR and ES for the beginning of 2017. We find that the Monte Carlo simulation-based estimates of VaR and ES are higher for the new MS DCS-EGARCH models than for MS Beta-t-EGARCH.

The remainder of this article is organized as follows. Section II presents the econometric framework. Section III describes the statistical inference. Section IV summarizes the empirical results. Section V presents the systematic risk analysis. Section VI concludes.

II. Econometric framework

For all econometric models, we use the daily log-return series $y_t = \ln(p_t/p_{t-1})$ for $t = 1, \ldots, T$, where p_t is the closing price of a financial asset for day t, and we use pre-sample data for p_0.

MS models

We use two-regime MS models (Hamilton 1989; Kim and Nelson 1999) for y_t, where the time-varying parameters of the model are driven by an underlying Markov chain $(s_1, \ldots, s_T)$ with $s_t \in \{1, 2\}$. We model the evolution of s_t by using the transition matrix:

$$P = \begin{bmatrix} \Pr(s_t = 1|s_{t-1} = 1) & \Pr(s_t = 2|s_{t-1} = 1) \\ \Pr(s_t = 1|s_{t-1} = 2) & \Pr(s_t = 2|s_{t-1} = 2) \end{bmatrix} = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix} \tag{1}$$

where p and q are the constant transition probability parameters, which are jointly estimated with all parameters of the score-driven regime-switching model. The Markov chain $(s_1, \ldots, s_T)$ is stationary with time-invariant probabilities $\pi^*(1) = \Pr(s_t = 1) = (1-q)/(2-p-q)$ and $\pi^*(2) = 1 - \pi^*(1)$ (the use of the stationary regime variable is a technical assumption that is needed for the statistical inference of score-driven regime-switching models). We also define the following filtered probabilities of regimes, in which we apply conditions that are related to the history of daily log-returns: $\pi_t(1) = \Pr(s_t = 1|y_1, \ldots, y_{t-1})$ and $\pi_t(2) = \Pr(s_t = 2|y_1, \ldots, y_{t-1})$, and $\tilde{\pi}_t(1) = \Pr(s_t = 1|y_1, \ldots, y_t)$. and $\tilde{\pi}_t(2) = \Pr(s_t = 2|y_1, \ldots, y_t)$. The two-regime DCS models of this article can be extended to multi-regime DCS models in a straightforward way.

MS DCS-EGARCH models

The general form of the MS DCS-EGARCH models that are estimated in this article is:

$$\begin{aligned} y_t &= \mu_t(s_t) + \nu_t(s_t) \\ &= \mu_t(s_t) + \exp[\lambda_t(s_t)]\varepsilon_t(s_t) \end{aligned} \tag{2}$$

where we use the Student's-t, Skew-Gen-t, EGB2 and NIG distributions with regime-dependent shape parameters for the error term $\varepsilon_t(s_t)$. The corresponding MS DCS-EGARCH models are denoted as MS Beta-t-EGARCH, MS Beta-Skew-Gen-t-EGARCH, MS EGB2-EGARCH and MS NIG-EGARCH, respectively. The use of the MS Beta-t-EGARCH model is suggested in the work of Blazsek and Ho (2017). To the best of our knowledge, the MS Beta-Skew-Gen-t-EGARCH, MS EGB2-EGARCH and MS

NIG-EGARCH models are new in the body of literature on DCS models. In addition to these MS DCS-EGARCH models, we also estimate their single-regime alternatives, in order to validate the consideration of the two regimes.

In Equation (2), the regime-dependent conditional location parameter $\mu_t(s_t)$ and the regime-dependent log of the conditional scale parameter $\lambda_t(s_t)$ are:

$$\mu_t(s_t) = c(s_t) \tag{3}$$

$$\begin{aligned}\lambda_t(s_t) = {} & \omega(s_t) + \beta(s_t)\lambda_{t-1}(s_t) \\ & + \alpha(s_t)u_{\lambda,t-1}(s_t) \\ & + \alpha^*(s_t)\text{sgn}[-v_{t-1}(s_t)][u_{\lambda,t-1}(s_t) + 1]\end{aligned} \tag{4}$$

respectively, where $\text{sgn}(x)$ is the signum function, and $c(s_t)$, $\omega(s_t)$, $\beta(s_t)$, $\alpha(s_t)$ and $\alpha^*(s_t)$ with $s_t = 1, 2$ are parameters to be estimated. We initialize $\lambda_t(s_t)$ by using the parameter $\lambda_0(s_t)$ with $s_t = 1, 2$. In Equation (4), the variables that update the log-scale parameter are defined as:

$$\lambda_{t-1}(s_t) = E[\lambda_{t-1}(s_{t-1})|y_1, \ldots, y_{t-1}, s_t] \tag{5}$$

$$u_{\lambda,t-1}(s_t) = E[u_{\lambda,t-1}(s_{t-1})|y_1, \ldots, y_{t-1}, s_t] \tag{6}$$

$$v_{t-1}(s_t) = E[v_{t-1}(s_{t-1})|y_1, \ldots, y_{t-1}, s_t] \tag{7}$$

where $u_{\lambda,t}(s_t) = \partial \ln f(y_t|y_1, \ldots, y_{t-1}, s_t)/\partial\lambda_t(s_t)$ within the expectation on the right-hand side of Equation (6) is the conditional score of the log-likelihood (LL) (hereinafter, score function) with respect to $\lambda_t(s_t)$; $f(y_t|y_1, \ldots, y_{t-1}, s_t)$ is the regime-dependent conditional density of y_t. The expectations in Equations (5) to (7) are computed with respect to s_{t-1} (see Appendix A). Therefore, the updating terms of Equation (4) depend only on the contemporaneous regime variable, which simplifies the statistical inference of the MS DCS-EGARCH models.

We present the mathematical details of each MS DCS-EGARCH model in Appendix B, where we present: (i) the regime-dependent probability distribution of $\varepsilon_t(s_t)$; (ii) the regime-dependent conditional mean $E(y_t|y_1, \ldots, y_{t-1}, s_t)$; (iii) the regime-dependent conditional volatility $\text{SD}(y_t|y_1, \ldots, y_{t-1}, s_t)$ (SD denotes standard deviation); (iv) the regime-dependent log conditional density $\ln f(y_t|y_1, \ldots, y_{t-1}, s_t)$; (v) the regime-dependent score function $u_{\lambda,t}(s_t)$.

It is noteworthy that, for all MS DCS-EGARCH specifications of this section, we assume that all parameters of Equations (2) to (4) may take different values in different regimes. We name those specifications as the unrestricted models.

III. Statistical inference

All models are estimated by using the ML method (Davidson and MacKinnon 2003):

$$\begin{aligned}\hat{\Theta}_{\text{ML}} &= \arg\max_{\Theta} \text{LL} \\ &= \arg\max_{\Theta} \sum_{T}^{t=1} \ln f(y_t|y_1, \ldots, y_{t-1})\end{aligned} \tag{8}$$

where Θ denotes the vector of parameters. For both single-regime and MS models, our criterion for parameter identification is convergence to the maximum LL at interior points of the parameter space, with convergence tolerance for the gradient 10^{-5}. We estimate the transformed values of several parameters, and the corresponding standard errors are computed by using the delta method. For all single-regime models, the ML estimation is robust for different start values of parameters. For all MS models, the start values of parameters are based on the parameter estimates of the corresponding single-regime alternative. For example, we use as start values $c(1) = 1.02 \times \hat{c}$ and $c(2) = 0.98 \times \hat{c}$, where $\hat{c}$ is the estimate from the single-regime model. For p and q, we use 98% and 99%, respectively, as start values.

For all MS models, the conditional density of y_t in Equation (8) is computed as the weighted average of the regime-dependent conditional densities (Appendix B), as follows:

$$\begin{aligned}f(y_t|y_1, \ldots, y_{t-1}) = {} & f(y_t|y_1, \ldots, y_{t-1}, s_t = 1)\pi_t(1) \\ & + f(y_t|y_1, \ldots, y_{t-1}, s_t = 2)\pi_t(2)\end{aligned} \tag{9}$$

where the weights are obtained by using the filtered probabilities, as follows:

$$\pi_t(1) = p\tilde{\pi}_{t-1}(1) + (1 - q)\tilde{\pi}_{t-1}(2) \tag{10}$$

$$\pi_t(2) = 1 - \pi_t(1) \tag{11}$$

$$\tilde{\pi}_t(1) = \frac{f(y_t|y_1,\ldots,y_{t-1},s_t=1)\pi_t(1)}{f(y_t|y_1,\ldots,y_{t-1},s_t=1)\pi_t(1)+f(y_t|y_1,\ldots,y_{t-1},s_t=2)\pi_t(2)} \tag{12}$$

$$\tilde{\pi}_t(2) = 1-\tilde{\pi}_t(1) \tag{13}$$

For the first period of the sample, we use the unconditional probabilities of the Markov chain $\tilde{\pi}_0(1)=\pi^*(1)$ and $\tilde{\pi}_0(2)=\pi^*(2)$ as initial probabilities (Kim and Nelson 1999).

Given the ML estimates of each MS model, the conditional mean and volatility of y_t are computed as the weighted average of the regime-dependent conditional means and volatilities of y_t (Appendix B), respectively, by using the filtered probabilities:

$$E(y_t|y_1,\ldots,y_{t-1}) = E(y_t|y_1,\ldots,y_{t-1},s_t=1)\pi_t(1) + E(y_t|y_1,\ldots,y_{t-1},s_t=2)\pi_t(2) \tag{14}$$

$$\mathrm{SD}(y_t|y_1,\ldots,y_{t-1}) = \mathrm{SD}(y_t|y_1,\ldots,y_{t-1},s_t=1)\pi_t(1) + \mathrm{SD}(y_t|y_1,\ldots,y_{t-1},s_t=2)\pi_t(2) \tag{15}$$

Given the ML estimates of each MS model, we make inferences on s_t by approximating the smoothed probability $\Pr(s_t|y_1,\ldots,y_T)$ (Kim and Nelson 1999). The smoothed probabilities of regimes $j=1,2$ are given by:

$$\Pr(s_t=j|y_1,\ldots,y_T) = \sum_{k=1,2}\Pr(s_t=j,s_{t+1}=k|y_1,\ldots,y_T) \tag{16}$$

where we use the approximation, as suggested by Kim and Nelson (1999):

$$\Pr(s_t=j,s_{t+1}=k|y_1,\ldots,y_T) \simeq \frac{\Pr(s_{t+1}=k|y_1,\ldots,y_T)\Pr(s_{t+1}=k|s_t=j)\tilde{\pi}_t(j)}{\pi_{t+1}(k)} \tag{17}$$

for $j=1,2$ and $k=1,2$. The smoothed probabilities of the regimes are computed recursively for $t=T,\ldots,1$. For $t=T$, we use the estimated values of p, q, $\pi_T(k)$ and $\tilde{\pi}_T(k)$.

IV. Empirical results

Data

We use time series data from the daily log-returns on the S&P 500 index for the period 1950–2016 (data source: Yahoo Finance, https://finance.yahoo.com/). In Table 1, we present the descriptive statistics of y_t. The negative skewness estimate indicates that the mass of the probability distribution of y_t is concentrated on the right side of the distribution, and the high excess kurtosis estimate suggests heavy tails of y_t. The negative correlation coefficient Corr (y_t^2, y_{t-1}) suggests that large volatility often follows large negative returns, which motivates the consideration of leverage effects in the DCS-EGARCH model. In Table 1, we also present the partial autocorrelation function (PACF) (Hamilton 1994) up to 30 lags.

ML estimation results for unrestricted models

In Table 2, we present the ML estimation results for the unrestricted models. The ML estimation procedure always converges effectively. We identify significant volatility dynamics, and we find two dynamic regimes of volatility. We define the high-volatility regime and the low-volatility regime by comparing the annualized unconditional volatilities of each regime. In Table 2, we indicate the high-volatility regime by bold numbers. In Figure 1, we present the evolution of the annualized conditional volatility $\mathrm{SD}(y_t|y_1,\ldots,y_{t-1})$. In the same figure, we also present the evolution of the smoothed probability of the high-volatility regime $\Pr(s_t|y_1,\ldots,y_T)$.

A common property of DCS models from the body of literature (Harvey 2013) is that extreme observations are discounted by the score function. We show that this property also holds for the new MS DCS-EGARCH models that are suggested in the present work. In Figure 2, we present the score function $u_{\lambda,t}(s_t)$ as a function of $\varepsilon_t(s_t)$, for all unrestricted MS DCS-EGARCH models and both regimes. In each panel of Figure 2, the thick line corresponds to the

Table 1. Descriptive statistics.

Start date	4 January 1950	PACF(1)	0.0273***	PACF(11)	−0.0142*	PACF(21)	−0.0181**
End date	30 December 2016	PACF(2)	0.0422***	PACF(12)	0.0280***	PACF(22)	−0.0028
Sample size (T)	16,858	PACF(3)	0.0028	PACF(13)	−0.0005	PACF(23)	−0.0007
Minimum	−0.2290	PACF(4)	−0.0082	PACF(14)	−0.0010	PACF(24)	0.0103
Maximum	0.1096	PACF(5)	−0.0129*	PACF(15)	−0.0130*	PACF(25)	−0.0117
Mean	0.0003	PACF(6)	−0.0055	PACF(16)	0.0325***	PACF(26)	−0.0214***
Standard deviation	0.0097	PACF(7)	−0.0190**	PACF(17)	−0.0045	PACF(27)	0.0192**
Skewness	−1.0110	PACF(8)	0.0105	PACF(18)	−0.0205***	PACF(28)	−0.0049
Excess kurtosis	27.1272	PACF(9)	−0.0108	PACF(19)	0.0016	PACF(29)	0.0236***
Corr(y_t^2,y_{t-1})	−0.0877	PACF(10)	0.0130*	PACF(20)	0.0106	PACF(30)	0.0064

Notes: The partial autocorrelation function (PACF) lag order is shown in parentheses. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table 2. Parameter estimates and model diagnostics (unrestricted models).

	Beta-t-EGARCH	Beta-Skew-Gen-t-EGARCH	EGB2-EGARCH	NIG-EGARCH
c	0.0004***(0.0001)	0.0008***(0.0001)	0.0010***(0.0001)	0.0010***(0.0001)
ω	−0.0647***(0.0067)	−0.0578***(0.0071)	−0.0641***(0.0079)	−0.0516***(0.0064)
α	0.0375***(0.0020)	0.0394***(0.0021)	0.0386***(0.0020)	0.0391***(0.0020)
α^*	0.0245***(0.0014)	0.0252***(0.0015)	0.0234***(0.0014)	0.0241***(0.0015)
β	0.9872***(0.0013)	0.9887***(0.0014)	0.9888***(0.0014)	0.9889***(0.0014)
λ_0	−5.1642***(0.2196)	−5.1399***(0.2255)	−5.6943***(0.2376)	−4.6250***(0.2360)
δ_1	1.6499***(0.0573)	−0.0503***(0.0089)	−0.2253***(0.0613)	0.7050***(0.0543)
δ_2	NA	2.0343***(0.0977)	−0.1160*(0.0650)	−0.0659***(0.0114)
δ_3	NA	0.5809***(0.0276)	NA	NA
	MS Beta-t-EGARCH	MS Beta-Skew-Gen-t-EGARCH	MS EGB2-EGARCH	MS NIG-EGARCH
$c(1)$	0.0006***(0.0001)	0.0008***(0.0001)	0.0010***(0.0002)	0.0010***(0.0002)
$c(2)$	0.0003***(0.0001)	0.0011***(0.0002)	0.0011***(0.0001)	0.0011***(0.0001)
$\omega(1)$	−0.0333***(0.0082)	−0.0668***(0.0094)	−0.0373***(0.0088)	−0.0308***(0.0072)
$\omega(2)$	−0.0908***(0.0088)	−0.0418***(0.0107)	−0.0859***(0.0125)	−0.0694***(0.0101)
$\alpha(1)$	0.0205***(0.0027)	0.0357***(0.0027)	0.0189***(0.0024)	0.0202***(0.0025)
$\alpha(2)$	0.0376***(0.0028)	0.0195***(0.0036)	0.0421***(0.0032)	0.0424***(0.0033)
$\alpha^*(1)$	0.0060***(0.0020)	0.0415***(0.0025)	0.0049***(0.0018)	0.0057***(0.0019)
$\alpha^*(2)$	0.0431***(0.0025)	0.0059**(0.0026)	0.0470***(0.0030)	0.0476***(0.0029)
$\beta(1)$	0.9934***(0.0016)	0.9874***(0.0019)	0.9932***(0.0016)	0.9932***(0.0016)
$\beta(2)$	0.9823***(0.0018)	0.9916***(0.0021)	0.9856***(0.0023)	0.9858***(0.0023)
$\lambda_0(1)$	−5.2796***(0.6417)	−5.0969***(0.5784)	−5.7423***(0.7056)	−4.7149***(0.7250)
$\lambda_0(2)$	−5.0633***(0.7179)	−5.3379***(0.8527)	−5.5871***(0.7669)	−4.5064***(0.8191)
$\delta_1(1)$	1.5646***(0.0832)	−0.0834***(0.0118)	−0.0585(0.0810)	0.7952***(0.0804)
$\delta_1(2)$	1.8350***(0.0974)	−0.0219(0.0191)	−0.2371***(0.0793)	0.7721***(0.0814)
$\delta_2(1)$	NA	2.4291***(0.2114)	−0.0067(0.0875)	−0.0317(0.0205)
$\delta_2(2)$	NA	1.6216***(0.1647)	−0.0557(0.0823)	−0.1068***(0.0154)
$\delta_3(1)$	NA	0.5657***(0.0402	NA	NA
$\delta_3(2)$	NA	0.6461***(0.0555	NA	NA
p	0.9982***(0.0006)	0.9968***(0.0010)	0.9971***(0.0008)	0.9977***(0.0007)
q	0.9982***(0.0007)	0.9950***(0.0014)	0.9973***(0.0009)	0.9977***(0.0008)
$E(\sigma_1)$	**11.74**%	9.03%	**12.03**%	**11.75**%
$E(\sigma_2)$	10.72%	**13.05**%	8.14%	8.17%

Notes: Standard errors are reported in parentheses. * and *** indicate significance at the 10% and 1% levels, respectively. $E(\sigma_1)$ and $E(\sigma_2)$ show the annualized unconditional volatility for each regime. Bold numbers indicate the high-volatility regime.

high-volatility regime, and the thin line corresponds to the low-volatility regime. Extreme observations are discounted for all MS DCS-EGARCH models. Moreover, the discounting of those observations is different for each DCS model and for each regime (Figure 2). For the new MS DCS-EGARCH models, an asymmetric discounting of extreme observations can be observed, with respect to the benchmark MS Beta-t-EGARCH model (Figure 2).

In Table 3, we present several model performance metrics that are used for both the single-regime and MS models. Firstly, we undertake a likelihood-ratio (LR) test for non-nested models (Vuong 1989). We denote the conditional density functions of the log-return of MS and single-regime models by using $f(y_t|y_1,\ldots,y_{t-1})$ and $g(y_t|y_1,\ldots,y_{t-1})$, respectively, and we define their difference as $d_t = \ln f(y_t|y_1,\ldots,y_{t-1}) - \ln g(y_t|y_1,\ldots,y_{t-1})$. We test whether LL of an MS model is superior to that of the corresponding single-regime alternative by estimating $d_t = c + \varepsilon_t$ with OLS-HAC (ordinary least squares heteroscedasticity and autocorrelation

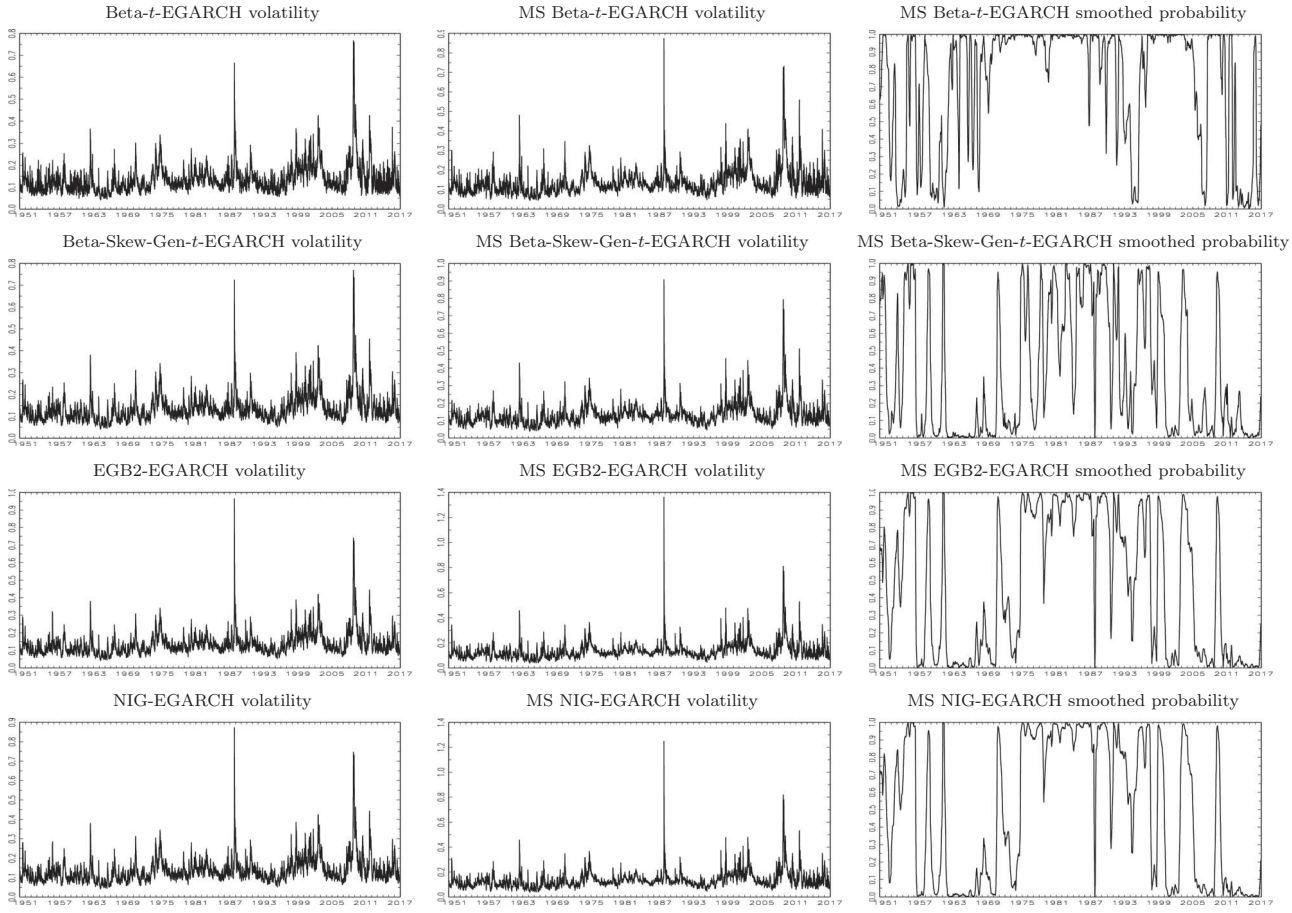


Figure 1. Annualized volatility of y_t and smoothed probability of the high-volatility regime (unrestricted models).

consistent) covariance estimation (Newey and West 1987). If c is significantly positive, then the MS model is superior to the single-regime alternative. For all cases, we find that the MS model is superior to the single-regime alternative (Table 3). Secondly, we use the following likelihood-based performance metrics: LL, Akaike information criterion (AIC), Bayesian information criterion (BIC) and Hannan-Quinn criterion (HQC) (Davidson and MacKinnon 2003) (Table 3). We find that the AIC, BIC and HQC-based performance of each MS model is superior to that of the corresponding single-regime alternative. We also find that the AIC, BIC and HQC-based performance of the new MS DCS-EGARCH models is superior to that of the MS Beta-t-EGARCH model.

ML estimation results for restricted models

For all MS models, we test whether the parameters are different in different regimes, by using robust tests of parameter differences that is based on the covariance matrix of the ML estimates. In Table 4, we present the corresponding test statistic for all unrestricted MS models. Those test results indicate whether some parameters have identical values in both regimes (e.g. $c(1) = c(2)$). These tests of parameter differences determine the restricted models.

In Table 5, we present the ML estimation results and the model diagnostics for the restricted models. The ML estimation procedure converges effectively for all specifications. We identify significant volatility dynamics, and we find two separate dynamic regimes of volatility (Table 5). In Figure 3, we present the evolution of annualized conditional volatility $\mathrm{SD}(y_t|y_1,\ldots,y_{t-1})$ and the evolution of the smoothed probability of the high-volatility regime $\Pr(s_t|y_1,\ldots,y_T)$, for the restricted models. In Figure 4, we present the score function $u_{\lambda,t}(s_t)$ for $s_t = 1, 2$, for all restricted MS DCS-EGARCH models. Similarly to the unrestricted models, we find different discounting of extreme observations for

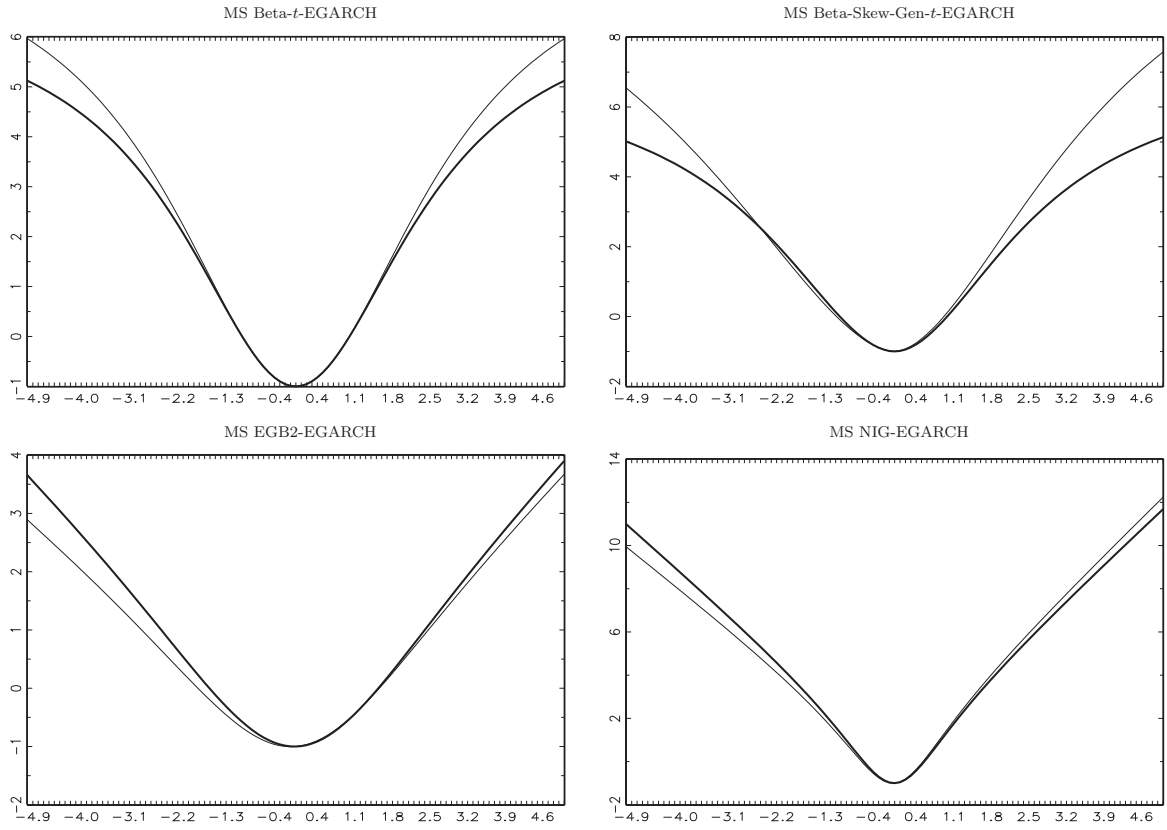


Figure 2. Score function $u_{\lambda,t}(s_t)$, as a function of $\varepsilon_t(s_t)$ (unrestricted models).

Notes: Each thick line corresponds to the high-volatility regime; each thin line corresponds to the low-volatility regime.

Table 3. Model performance (unrestricted models).

	SR LL(rank)	MS LL(rank)	LR
Beta-*t*-EGARCH	3.4402(3)	**3.4433**(4)	0.0040***(0.0009)
Beta-Skew-Gen-*t*-EGARCH	3.4415(1)	**3.4452**(1)	0.0038***(0.0007)
EGB2-EGARCH	3.4398(4)	**3.4439**(3)	0.0040***(0.0007)
NIG-EGARCH	3.4402(2)	**3.4441**(2)	0.0039***(0.0007)

	SR AIC(rank)	MS AIC(rank)	SR BIC(rank)	MS BIC(rank)	SR HQC(rank)	MS HQC(rank)
Beta-*t*-EGARCH	− 6.8795(3)	− **6.8847**(4)	− 6.8763(2)	− **6.8774**(3)	− 6.8784(2)	− **6.8823**(4)
Beta-Skew-Gen-*t*-EGARCH	− 6.8819(1)	− **6.8881**(1)	− 6.8777(1)	− **6.8789**(1)	− 6.8805(1)	− **6.8851**(1)
EGB2-EGARCH	− 6.8787(4)	− **6.8856**(3)	− 6.8751(4)	− **6.8774**(4)	− 6.8775(4)	− **6.8829**(3)
NIG-EGARCH	− 6.8795(2)	− **6.8860**(2)	− 6.8758(3)	− **6.8778**(2)	− 6.8783(3)	− **6.8833**(2)

Notes: *** indicates significance at the 1% level. Single regime (SR); Markov-switching (MS); log-likelihood (LL); likelihood-ratio test (LR); Akaike information criterion (AIC); Bayesian information criterion (BIC); Hannan–Quinn criterion (HQC). For the LR test, we estimate $d_t = c + \varepsilon_t$ by using OLS-HAC, where d_t is the difference between the log-densities of MS and SR for observation t. We report the robust estimate of c, with standard errors in parentheses. A significantly positive c indicates that the LL of MS is superior to that of SR. Better model performance is indicated by bold numbers.

Table 4. Test statistics of differences in parameters (unrestricted models).

	MS Beta-*t*-EGARCH	MS Beta-Skew-Gen-*t*-EGARCH	MS EGB2-EGARCH	MS NIG-EGARCH
$c(1) - c(2)$	3.6373***(0.0003)	− 1.3355(0.1817)	− 0.2426(0.8083)	− 0.4443(0.6569)
$\omega(1) - \omega(2)$	4.7812***(0.0000)	− 1.7592*(0.0786)	3.1862***(0.0014)	3.1191***(0.0018)
$\alpha(1) - \alpha(2)$	− 4.3959***(0.0000)	3.5967***(0.0003)	− 5.7818***(0.0000)	− 5.3943***(0.0000)
$\alpha^*(1) - \alpha^*(2)$	− 11.6170***(0.0000)	9.9166***(0.0000)	− 12.1009***(0.0000)	− 12.0841***(0.0000)
$\beta(1) - \beta(2)$	4.6197***(0.0000)	− 1.4749(0.1402)	2.7230***(0.0065)	2.6247***(0.0087)
$\lambda_0(1) - \lambda_0(2)$	− 0.2246(0.8223)	0.2339(0.8151)	− 0.1489(0.8817)	− 0.1907(0.8488)
$\delta_1(1) - \delta_1(2)$	− 2.1099**(0.0349)	− 2.7450***(0.0061)	1.5765(0.1149)	0.2016(0.8402)
$\delta_2(1) - \delta_2(2)$	NA	3.0132***(0.0026)	0.4085(0.6829)	2.9258***(0.0034)
$\delta_3(1) - \delta_3(2)$	NA	− 1.1733(0.2407)	NA	NA

Notes: *p*-values are reported in parentheses. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table 5. Parameter estimates and model diagnostics (restricted models).

	MS Beta-*t*-EGARCH	MS Beta-Skew-Gen-*t*-EGARCH	MS EGB2-EGARCH	MS NIG-EGARCH
$c(1)$	0.0006***(0.0001)	0.0009***(0.0001)	0.0010***(0.0001)	0.0011***(0.0001)
$c(2)$	0.0003***(0.0001)	$c(1)$	$c(1)$	$c(1)$
$\omega(1)$	− 0.0335***(0.0082)	− 0.0553***(0.0066)	− 0.0341***(0.0081)	− 0.0312***(0.0071)
$\omega(2)$	− 0.0906***(0.0088)	− 0.0513***(0.0066)	− 0.1000***(0.0124)	− 0.0689***(0.0094)
$\alpha(1)$	0.0206***(0.0027)	0.0355***(0.0027)	0.0221***(0.0023)	0.0203***(0.0025)
$\alpha(2)$	0.0375***(0.0028)	0.0206***(0.0035)	0.0440***(0.0034)	0.0417***(0.0032)
$\alpha^*(1)$	0.0061***(0.0020)	0.0414***(0.0025)	0.0079***(0.0017)	0.0056***(0.0019)
$\alpha^*(2)$	0.0431***(0.0025)	0.0060**(0.0026)	0.0485***(0.0030)	0.0471***(0.0029)
$\beta(1)$	0.9934***(0.0016)	0.9897***(0.0013)	0.9939***(0.0015)	0.9931***(0.0016)
$\beta(2)$	0.9824***(0.0018)	$\beta(1)()$	0.9831***(0.0022)	0.9859***(0.0021)
$\lambda_0(1)$	− 5.2480***(0.1886)	− 5.3177***(0.1868)	− 5.7372***(0.1949)	− 4.7055***(0.1927)
$\lambda_0(2)$	$\lambda_0(1)$	$\lambda_0(1)$	$\lambda_0(1)$	$\lambda_0(1)$
$\delta_1(1)$	1.5642***(0.0831)	− 0.0922***(0.0101)	− 0.1738***(0.0641)	0.7827***(0.0595)
$\delta_1(2)$	1.8355***(0.0975)	− 0.0110(0.0122)	$\delta_1(1)$	$\delta_1(1)$
$\delta_2(1)$	NA	2.3170***(0.1683)	− 0.0629(0.0672)	− 0.0379***(0.0126)
$\delta_2(2)$	NA	1.7932***(0.1246)	$\delta_2(1)$	− 0.1030***(0.0128)
$\delta_3(1)$	NA	0.5947***(0.0305)	NA	NA
$\delta_3(2)$	NA	$\delta_3(1)$	NA	NA
p	0.9982***(0.0006)	0.9968***(0.0010)	0.9984***(0.0005)	0.9977***(0.0007)
q	0.9982***(0.0007)	0.9954***(0.0013)	0.9980***(0.0007)	0.9977***(0.0008)
$E(\sigma_1)$	**11.75**%	8.30%	**11.66**%	**11.76**%
$E(\sigma_2)$	10.72%	**12.86**%	8.41%	8.25%

Notes: Standard errors are reported in parentheses. ** and *** indicate significance at the 5% and 1% levels, respectively. $E(\sigma_1)$ and $E(\sigma_2)$ show the annualized unconditional volatility for each regime. Bold numbers indicate the high-volatility regime.

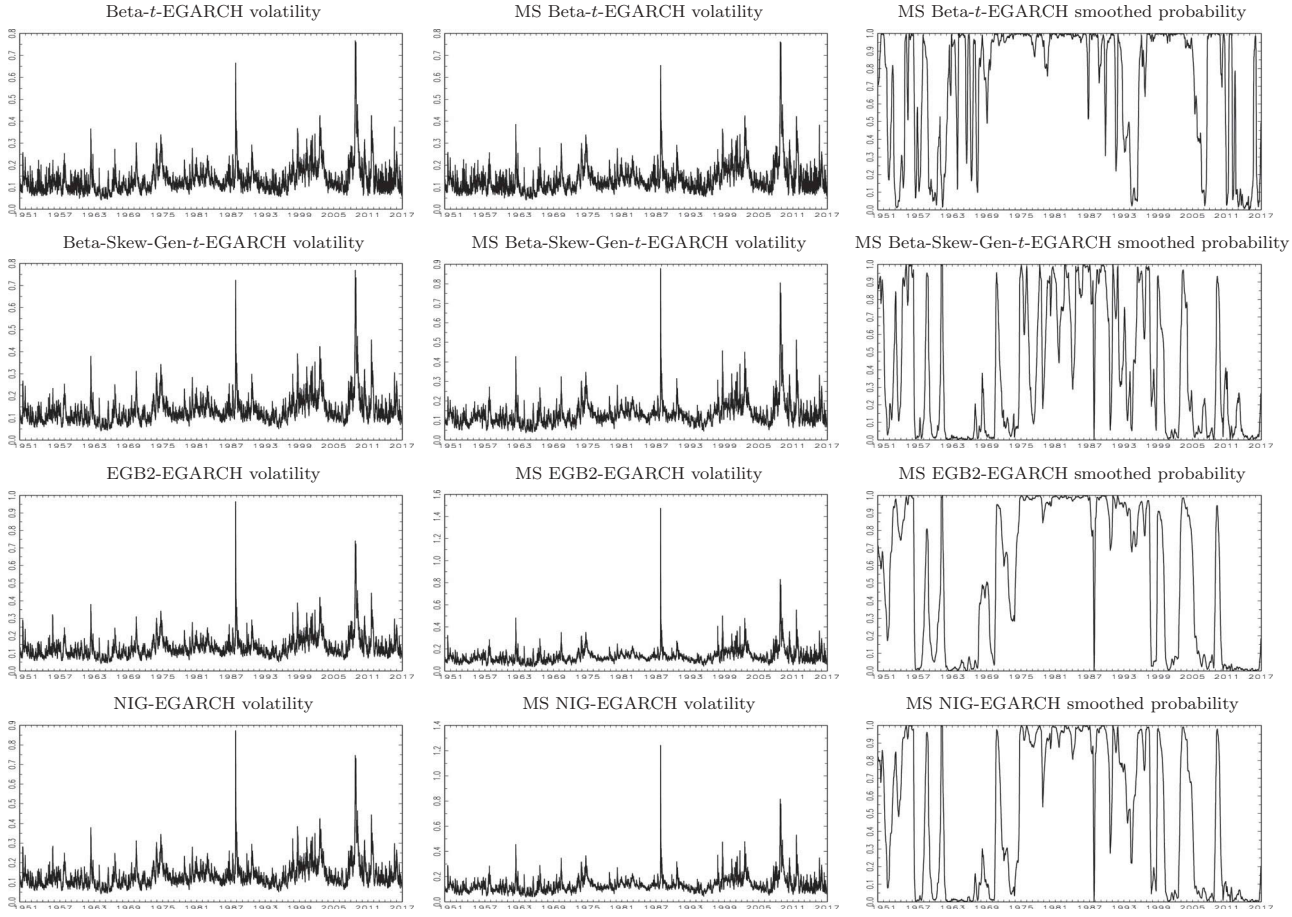


Figure 3. Annualized volatility of y_t and smoothed probability of the high-volatility regime (restricted models).

Notes: For the single-regime models we repeat Figure 1, for the purpose of model comparison.

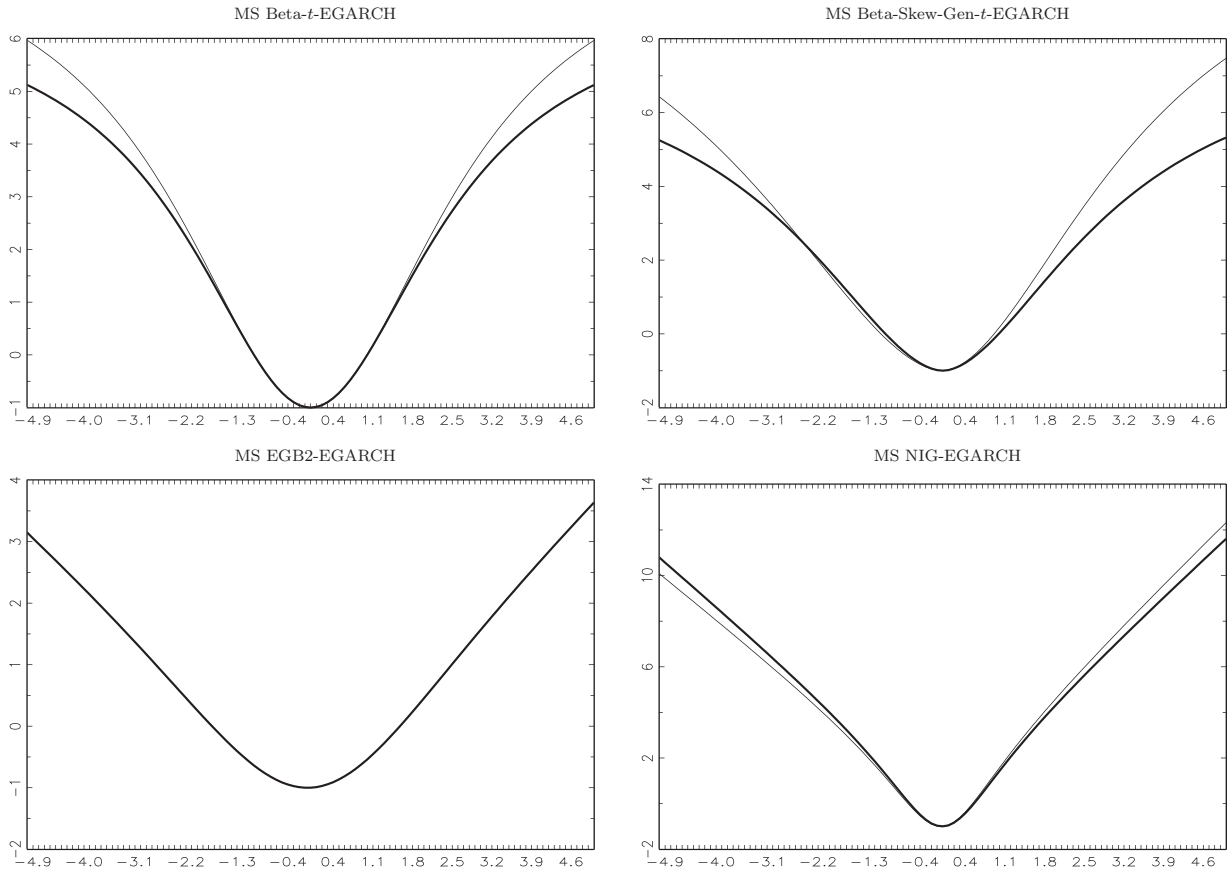


Figure 4. Score function $u_{\lambda,t}(s_t)$, as a function of $\varepsilon_t(s_t)$ (restricted models).

Notes: Each thick line corresponds to the high-volatility regime; each thin line corresponds to the low-volatility regime.

different DCS models, and we also find different discounting in each regime for the MS Beta-*t*-EGARCH, MS Beta-Skew-*t*-EGARCH and MS NIG-EGARCH models (for the restricted MS EGB2-EGARCH model, the score functions are identical in both regimes).

In Table 6, we present several model performance metrics for the restricted MS models: We present the LR test results and also the AIC, BIC and HQC metrics. All results indicate that the MS model is superior to the single-regime alternative. Moreover, the results also show that the new MS DCS-EGARCH models are superior to the MS Beta-*t*-EGARCH model.

V. Systematic risk analysis

In this section, we forecast VaR (Jorion 2006) and ES (Acharya et al., 2012; Acharya et al. 2017) of one long position on the S&P 500, by using the restricted MS DCS-EGARCH models. As aforementioned, this analysis is useful for investors of (i) well-diversified US equity portfolios; (ii) S&P 500 futures and options traded at Chicago Mercantile Exchange Globex; (iii) ETFs related to the S&P 500. For practitioners, $\text{VaR}_{\tilde{\alpha}}$ indicates that the loss on the S&P 500 position, over the following T_h days, will be higher than $\text{VaR}_{\tilde{\alpha}}$ with probability $\tilde{\alpha}$. Furthermore, $\text{ES}_{\tilde{\alpha}}$ indicates the average loss on the S&P 500 position over the following T_h days, given that the loss already exceeds $\text{VaR}_{\tilde{\alpha}}$. In this article, both VaR and ES are calculated in terms of return on the investment. We use the alternative investment time horizons $T_h = 1, 5, 10$ and 30 days, and the alternative confidence levels $1 - \tilde{\alpha} = 90\%, 95\%$ and 99%. The estimation window of MS DCS-EGARCH for VaR and ES is from 1950 to 2016 (i.e. the full data window of this study; see Table 1). Thus, each investment time horizon corresponds to the first T_h trading days of 2017. Mathematically, VaR and ES are defined as follows:

$$\Pr\left(\sum_{s=1}^{T_h} y_{t+s} < -\text{VaR}_{\tilde{\alpha}}\right) = \tilde{\alpha} \tag{18}$$

Table 6. Model performance (restricted models).

	SR LL(rank)	MS LL(rank)	LR
Beta-*t*-EGARCH	3.4402(3)	**3.4433**(3)	0.0032***(0.0007)
Beta-Skew-Gen-*t*-EGARCH	3.4415(1)	**3.4451**(1)	0.0037***(0.0007)
EGB2-EGARCH	3.4398(4)	**3.4431**(4)	0.0032***(0.0007)
NIG-EGARCH	3.4402(2)	**3.4441**(2)	0.0039***(0.0007)

	SR AIC(rank)	MS AIC(rank)	SR BIC(rank)	MS BIC(rank)	SR HQC(rank)	MS HQC(rank)
Beta-*t*-EGARCH	− 6.8795(3)	− **6.8848**(3)	− 6.8763(2)	− **6.8780**(4)	− 6.8784(2)	− **6.8826**(3)
Beta-Skew-Gen-*t*-EGARCH	− 6.8819(1)	− **6.8884**(1)	− 6.8777(1)	− **6.8810**(1)	− 6.8805(1)	− **6.8859**(1)
EGB2-EGARCH	− 6.8787(4)	− **6.8845**(4)	− 6.8751(4)	− **6.8780**(3)	− 6.8775(4)	− **6.8823**(4)
NIG-EGARCH	− 6.8795(2)	− **6.8864**(2)	− 6.8758(3)	− **6.8795**(2)	− 6.8783(3)	− **6.8841**(2)

Notes: *** indicates significance at the 1% level. Single regime (SR); Markov-switching (MS); log-likelihood (LL); likelihood-ratio test (LR); Akaike information criterion (AIC); Bayesian information criterion (BIC); Hannan–Quinn criterion (HQC). For the LR test, we estimate $d_t = c + \varepsilon_t$ by using OLS-HAC, where d_t is the difference between the log-densities of MS and SR for observation t. We report the robust estimate of c, with standard errors in parentheses. A significantly positive c indicates that the LL of MS is superior to that of SR. Better model performance is indicated by bold numbers. For the single-regime models, we repeat the metrics of Table 3, for the purpose of model comparison.

and

$$\mathrm{ES}_{\tilde{\alpha}} = -E\left(\sum_{s=1}^{T_h} y_{t+s} \middle| \sum_{s=1}^{T_h} y_{t+s} \leq -\mathrm{VaR}_{\tilde{\alpha}}\right), \qquad (19)$$

respectively. In Equations (18) and (19), we compute the total return on the investment for T_h, by using the additive property of the log-return (i.e. the total log-return equals to the sum of the consecutive daily log-returns). We estimate both VaR and ES, by using the Monte Carlo simulation method: We simulate 100,000 trajectories from y_t for each time horizon T_h. Then, $\mathrm{VaR}_{\tilde{\alpha}}$ is estimated by using the $(100 \times \tilde{\alpha})$-th percentile of the total return on the investment. Furthermore, ES is estimated by using the average of those simulated total returns on the investment that are lower than, or equal to, $-\mathrm{VaR}_{\tilde{\alpha}}$. In Table 7, we present the VaR and ES estimates for all MS DCS-EGARCH models. Those results suggest that both VaR and ES are higher for the new MS DCS-EGARCH models than for the MS Beta-*t*-EGARCH model.

Table 7. Value-at-risk (VaR) and expected shortfall (ES) estimates (restricted models).

	T_h = 1day	T_h = 5days	T_h = 10days	T_h = 30days
MS Beta-*t*-EGARCH				
$\mathrm{VaR}_{1\%}$	1.1253%	2.3812%	3.4096%	5.5951%
$\mathrm{VaR}_{5\%}$	0.7204%	1.5365%	2.1434%	3.2755%
$\mathrm{VaR}_{10\%}$	0.5336%	1.1288%	1.5321%	2.1242%
$\mathrm{ES}_{1\%}$	1.3879%	2.9108%	4.1359%	6.9217%
$\mathrm{ES}_{5\%}$	0.9757%	2.0741%	2.9347%	4.7063%
$\mathrm{ES}_{10\%}$	0.7963%	1.6940%	2.3684%	3.6736%
MS Beta-Skew-Gen-*t*-EGARCH				
$\mathrm{VaR}_{1\%}$	1.1675%	2.6814%	4.1172%	8.0506%
$\mathrm{VaR}_{5\%}$	0.7586%	1.8002%	2.7387%	5.2700%
$\mathrm{VaR}_{10\%}$	0.5701%	1.3747%	2.0694%	3.9413%
$\mathrm{ES}_{1\%}$	1.4217%	3.2312%	4.9045%	9.6597%
$\mathrm{ES}_{5\%}$	1.0159%	2.3557%	3.5822%	7.0073%
$\mathrm{ES}_{10\%}$	0.8355%	1.9622%	2.9757%	5.7712%
MS EGB2-EGARCH				
$\mathrm{VaR}_{1\%}$	1.1881%	2.6167%	3.8080%	6.8868%
$\mathrm{VaR}_{5\%}$	0.7454%	1.7188%	2.4687%	4.3360%
$\mathrm{VaR}_{10\%}$	0.5511%	1.2894%	1.8236%	3.0580%
$\mathrm{ES}_{1\%}$	1.4588%	3.1248%	4.6049%	8.3900%
$\mathrm{ES}_{5\%}$	1.0205%	2.2684%	3.3017%	5.9326%
$\mathrm{ES}_{10\%}$	0.8293%	1.8751%	2.7089%	4.7832%
MS NIG-EGARCH				
$\mathrm{VaR}_{1\%}$	0.8540%	2.5083%	3.8516%	7.0226%
$\mathrm{VaR}_{5\%}$	0.7079%	1.7487%	2.5309%	4.3690%
$\mathrm{VaR}_{10\%}$	0.6020%	1.3287%	1.8561%	3.0353%
$\mathrm{ES}_{1\%}$	0.8952%	2.8867%	4.5172%	8.6326%
$\mathrm{ES}_{5\%}$	0.7947%	2.2174%	3.3227%	6.0362%
$\mathrm{ES}_{10\%}$	0.7236%	1.8681%	2.7412%	4.8360%

Notes: We estimate VaR and ES, by using the Monte Carlo simulation method with 100,000 trajectories simulated from y_t.

VI. Conclusion

We have suggested the use of new MS DCS-EGARCH models, to be applied by practitioners for forecasting VaR and ES in systematic risk analysis. We have used daily log-return data from the S&P 500 for the period 1950–2016. The new MS DCS-EGARCH models of this article are alternatives to the recent MS Beta-*t*-EGARCH model having the symmetric Student's *t* distribution with regime-switching degrees of freedom parameter for the error term. For the new MS DCS-EGARCH models, we have used flexible asymmetric probability distributions for the error term: Skew-Gen-*t*, EGB2 and NIG distributions. Thus, we have introduced the MS Beta-Skew-Gen-*t*-EGARCH, MS EGB2-EGARCH and MS NIG-EGARCH models, respectively. We have compared the new MS DCS-EGARCH models with the MS Beta-*t*-EGARCH model, by considering leverage effects in all DCS-EGARCH models. For all MS models, we have used two switching regimes and a constant transition probability matrix.

We have considered both unrestricted and restricted versions of each MS DCS-EGARCH model. For the unrestricted MS models, all parameters may have taken different values in different regimes. For the restricted MS models, some parameters have taken the same values in both regimes. We have defined the restricted models, by using robust statistical tests of parameter differences. For all MS DCS-EGARCH models, we have identified high and low-volatility periods for the S&P 500. We have found that the MS model is always superior to the single-regime alternative. We have found that the new MS DCS-EGARCH models are superior to the MS Beta-*t*-EGARCH model. Finally, as a practical application, we have performed a systematic risk analysis by forecasting VaR and ES for 2017. We have found that the VaR and ES metrics are higher for the new and more parsimonious MS DCS-EGARCH models than for the MS Beta-*t*-EGARCH model.

Acknowledgments

The authors are thankful to Matthew Copley and Jason Jones.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

Szabolcs Blazsek acknowledges funding from the School of Business of Universidad Francisco Marroquín.

References

Acharya, V., R. Engle, and M. Richardson. 2012. "Capital Shortfall: A New Approach to Ranking and Regulating Systemic Risks." *American Economic Review* 102 (3): 59–64. doi:10.1257/aer.102.3.59.

Acharya, V. V., L. H. Pedersen, T. Philippon, and M. Richardson. 2017. "Measuring Systemic Risk." *The Review of Financial Studies* 30 (1): 2–47. doi:10.1093/rfs/hhw088.

Bahr, K. M., and W. E. Maas. 2014. "International Equity Returns: Absolute Vs. Dollar Adjusted." *Journal of Accounting and Finance* 14 (6): 98–112. ISSN: 2158-3625.

Barndorff-Nielsen, O., and C. Halgreen. 1977. "Infinite Divisibility of the Hyperbolic and Generalized Inverse Gaussian Distributions." *Probability Theory and Related Fields* 38 (4): 309–311. doi:10.1007/bf00533162.

Black, F. Studies of Stock Price Volatility Changes. Proceedings of the 1976 meetings of the American Statistical Association, 1976.

Blazsek, S., and H. C. Ho. 2017. "Markov Regime-Switching Beta-T-Egarch." *Applied Economics* 49 (47): 4793–4805. doi:10.1080/00036846.2017.1293794.

Caivano, M., and A. C. Harvey. 2014. "Time-Series Models with an Egb2 Conditional Distribution." *Journal of Time Series Analysis* 35 (6): 558–571. doi:10.1111/jtsa.12081.

Davidson, R., and J. G. MacKinnon. 2003. *Econometric Theory and Methods*. New York: Oxford University Press.

Gray, S. F. 1996. "Modeling the Conditional Distribution of Interest Rates as a Regime-Switching Process." *Journal of Financial Economics* 42 (1): 27–62. doi:10.1016/0304-405X(96)00875-6.

Hamilton, J. D. 1989. "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica* 57 (2): 357–384. doi:10.2307/1912559.

Hamilton, J. D. 1994. *Time Series Analysis*. Princeton: Princeton University Press.

Harvey, A. C. 2013. *Dynamic Models for Volatility and Heavy Tails*. Cambridge: Cambridge University Press.

Harvey, A. C., and T. Chakravarty. Beta-T-(E)Garch. *Cambridge Working Papers in Economics 0840, Cambridge*, Cambridge, 2008. http://www.econ.cam.ac.uk/research/repec/cam/pdf/cwpe0840.pdf.

Harvey, A. C., and R. J. Lange. 2017. "Volatility Modeling with a Generalized T-Distribution." *Journal of Time Series Analysis* 38 (2): 175–190. doi:10.1111/jtsa.12224.

Jorion, P. 2006. *Value at Risk: The New Benchmark for Managing Financial Risk*. 3rd ed. New York: McGraw-Hill.

Kim, C. J., and C. R. Nelson. 1999. *State-Space Models with Regime Switching*. Cambridge: MIT Press.

Klaassen, F. 2002. "Improving Garch Volatility Forecasts with Regime-Switching Garch." *Empirical Economics* 27 (2): 363–394. doi:10.1007/s001810100100.

Newey, K., and K. D. West. 1987. "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55 (3): 703–738. doi:10.2307/1913610.

Schwartz, A. L., B. F. Van Ness, and R. A. Van Ness. 2004. "Clustering in the Futures Market: Evidence from S&P 500 Futures Contracts." *The Journal of Futures Markets* 24 (5): 413–428. doi:10.1002/fut.10129.

Vuong, Q. H. 1989. "Likelihood Ratio Tests for Model Selection and Non-Nested Hypotheses." *Econometrica* 57 (2): 307–333. doi:10.2307/1912557.

Xu, L., and X. Yin. 2017. "Exchange Traded Funds and Stock Market Volatility." *International Review of Finance* 17 (4): 525–560. doi:10.1111/irfi2121.

Appendix A Updating terms of MS DCS-EGARCH models

For all MS models, the conditional distribution of y_t depends on s_t, due to Equations (5) to (7) in which s_{t-1} is integrated out. This idea was suggested by Gray (1996) for MS-GARCH, who used $\lambda_{t-1} = E[\lambda_{t-1}(s_{t-1})|y_1,\ldots,y_{t-2}]$ to integrate out s_{t-1} from $\lambda_{t-1}(s_{t-1})$. Klaassen (2002) extended the work of Klaassen (2002) for MS-GARCH, and used $\lambda_{t-1}(s_t) = E[\lambda_{t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t]$.

We compute Equation (5) according to Klaassen (2002), as:

$$\begin{aligned}\lambda_{t-1}(s_t=1) &= E[\lambda_{t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=1]\\ &= \lambda_{t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=1)\\ &+ \lambda_{t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=1)\end{aligned} \tag{A.1}$$

$$\begin{aligned}\lambda_{t-1}(s_t=2) &= E[\lambda_{t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=2]\\ &= \lambda_{t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=2)\\ &+ \lambda_{t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=2)\end{aligned} \tag{A.2}$$

We compute Equation (6) as:

$$\begin{aligned}u_{\lambda,t-1}(s_t=1) &= E[u_{\lambda,t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=1]\\ &= u_{\lambda,t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=1)\\ &+ u_{\lambda,t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=1)\end{aligned} \tag{A.3}$$

$$\begin{aligned}u_{\lambda,t-1}(s_t=2) &= E[u_{\lambda,t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=2]\\ &= u_{\lambda,t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=2)\\ &+ u_{\lambda,t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=2)\end{aligned} \tag{A.4}$$

Furthermore, we compute Equation (7) as:

$$\begin{aligned}v_{t-1}(s_t=1) &= E[v_{t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=1]\\ &= v_{t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=1)\\ &+ v_{t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=1)\end{aligned} \tag{A.5}$$

$$\begin{aligned}v_{t-1}(s_t=2) &= E[v_{t-1}(s_{t-1})|y_1,\ldots,y_{t-1},s_t=2]\\ &= v_{t-1}(s_{t-1}=1)\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=2)\\ &+ v_{t-1}(s_{t-1}=2)\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=2)\end{aligned} \tag{A.6}$$

The probabilities of Equations (A.1) to (A.6) are given by:

$$\begin{aligned}&\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=1)\\ &= \frac{p\tilde{\pi}_{t-1}(1)}{p\tilde{\pi}_{t-1}(1)+(1-q)\tilde{\pi}_{t-1}(2)}\end{aligned} \tag{A.7}$$

$$\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=1) = 1-\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=1) \tag{A.8}$$

$$\begin{aligned}&\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=2)\\ &= \frac{(1-p)\tilde{\pi}_{t-1}(1)}{(1-p)\tilde{\pi}_{t-1}(1)+q\tilde{\pi}_{t-1}(2)}\end{aligned} \tag{A.9}$$

$$\Pr(s_{t-1}=2|y_1,\ldots,y_{t-1},s_t=2) = 1-\Pr(s_{t-1}=1|y_1,\ldots,y_{t-1},s_t=2) \tag{A.10}$$

and for the first period we use $\tilde{\pi}_0(1) = \pi^*(1)$ and $\tilde{\pi}_0(2) = \pi^*(2)$, as initial probabilities.

Appendix B MS DCS-EGARCH models

MS Beta-t-EGARCH

(i) The error term has the Student's-t distribution $\varepsilon_t(s_t)\sim t\{0,1,\exp[\delta_1(s_t)]+2\}$, where $\delta_1(s_t)$ with $s_t = 1,2$ is a shape parameter. The parameter $\delta_1(s_t)$ influences the tail-heaviness of $\varepsilon_t(s_t)$. The degrees of freedom parameter $\{\exp[\delta_1(s_t)]+2\}$ is greater than two, hence, the conditional variance of y_t is finite. (ii) The regime-dependent conditional mean of y_t is $\mu_t(s_t)$. (iii) The regime-dependent conditional volatility of y_t is $\exp[\lambda_t(s_t)]\{1+2\exp[-\delta_1(s_t)]\}^{1/2}$. (iv) The regime-dependent log conditional density of y_t is:

$$\begin{aligned}&\ln f(y_t|y_1,\ldots,y_{t-1},s_t)\\ &= \ln\Gamma\left\{\frac{\exp[\delta_1(s_t)]+3}{2}\right\} - \ln\Gamma\left\{\frac{\exp[\delta_1(s_t)]+2}{2}\right\}\\ &- \frac{\ln(\pi)+\ln\{\exp[\delta_1(s_t)]+2\}}{2} - \lambda_t(s_t)\\ &- \frac{\exp[\delta_1(s_t)]+3}{2}\ln\left\{1+\frac{\varepsilon_t^2(s_t)}{\exp[\delta_1(s_t)]+2}\right\}\end{aligned}, \tag{B.1}$$

where $\Gamma(x)$ is the gamma function. (v) The regime-dependent score function is given by:

$$u_{\lambda,t}(s_t) = \frac{\{\exp[\delta_1(s_t)]+3\}\varepsilon_t^2(s_t)}{\exp[\delta_1(s_t)]+2+\varepsilon_t^2(s_t)} - 1 \tag{B.2}$$

MS Beta-Skew-Gen-t-EGARCH

(i) The error term has the skewed generalized t distribution:

$$\begin{aligned}&\varepsilon_t(s_t)\sim \text{Skew}-\text{Gen}\\ &- t\{0,1,\tanh[\delta_1(s_t)],\exp[\delta_2(s_t)]+2,\exp[\delta_3(s_t)]\}\end{aligned} \tag{B.3}$$

where $\tanh(x)$ is the hyperbolic tangent function. Furthermore, $\delta_1(s_t)$, $\delta_2(s_t)$ and $\delta_3(s_t)$ with $s_t = 1,2$ are shape parameters that influence asymmetry, tail-heaviness and peakedness of $\varepsilon_t(s_t)$, respectively. The degrees of freedom parameter $\{\exp[\delta_1(s_t)]+2\}$ is greater than two, hence, the conditional

variance of y_t is finite. The asymmetry parameter is given by $\tanh(\tau_t) \in (-1, 1)$, as required for Skew-Gen-t.

(ii) The regime-dependent conditional mean of y_t is:

$$\mu_t(s_t) + 2\exp[\lambda_t(s_t)]\tanh[\delta_1(s_t)]\{\exp[\delta_2(s_t)] + 2\}^{\exp[-\delta_3(s_t)]} \times \frac{B\left\{\frac{2}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]+1}{\exp[\delta_3(s_t)]}\right\}}{B\left\{\frac{1}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]+2}{\exp[\delta_3(s_t)]}\right\}} \tag{B.4}$$

where $B(x, y)$ is the Beta function.

(iii) The regime-dependent conditional volatility of y_t is:

$$\exp[\lambda_t(s_t)]\{\exp[\delta_2(s_t)] + 2\}^{\exp[-\delta_3(s_t)]} \times \left\{ \frac{\{3\tanh^2[\delta_1(s_t)]+1\}B\left\{\frac{3}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]}{\exp[\delta_3(s_t)]}\right\}}{B\left[\frac{1}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]+2}{\exp[\delta_3(s_t)]}\right]} - \frac{4\tanh^2[\delta_1(s_t)]B^2\left\{\frac{2}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]+1}{\exp[\delta_3(s_t)]}\right\}}{B^2\left\{\frac{1}{\exp[\delta_3(s_t)]}, \frac{\exp[\delta_2(s_t)]+2}{\exp[\delta_3(s_t)]}\right\}} \right\}^{1/2} \tag{B.5}$$

(iv) The regime-dependent log conditional density of y_t is:

$$\begin{aligned}\ln f(y_t|y_1, \ldots, y_{t-1}, s_t) = {} & \delta_3(s_t) - \lambda_t(s_t) - \ln(2) \\ & - \frac{\ln\{\exp[\delta_2(s_t)\} + 2]}{\exp[\delta_3(s_t)]} \\ & - \ln\Gamma\left\{\frac{\exp[\delta_2(s_t)] + 2}{\exp[\delta_3(s_t)]}\right\} \\ & - \ln\Gamma\{\exp[-\delta_3(s_t)]\} \\ & + \ln\Gamma\left\{\frac{\exp[\delta_2(s_t)] + 3}{\exp[\delta_3(s_t)]}\right\} \\ & - \frac{\exp[\delta_2(s_t)] + 3}{\exp[\delta_3(s_t)]} \times \\ & \ln\left\{1 + \frac{|\varepsilon_t(s_t)|^{\exp[\delta_3(s_t)]}}{\{1 + \tanh[\delta_1(s_t)]\mathrm{sgn}[\varepsilon_t(s_t)]\}^{\exp[\delta_3(s_t)]} \times \{\exp[\delta_2(s_t)] + 2\}}\right\}\end{aligned} \tag{B.6}$$

(v) The regime-dependent score function is given by:

$$u_{\lambda,t}(s_t) = \frac{|\varepsilon_t(s_t)|^{\exp[\delta_3(s_t)]}\{\exp[\delta_2(s_t)] + 3\}}{|\varepsilon_t(s_t)|^{\exp[\delta_3(s_t)]} + \{1 + \tanh[\delta_1(s_t)]\mathrm{sgn}[\varepsilon_t(s_t)]\}^{\exp[\delta_3(s_t)]}\{\exp[\delta_2(s_t)] + 2\}} - 1 \tag{B.7}$$

MS EGB2-EGARCH

(i) The error term has the exponential generalized beta distribution of the second kind:

$$\varepsilon_t(s_t) \sim \mathrm{EGB2}\{0, 1, \exp[\delta_1(s_t)], \exp[\delta_2(s_t)]\} \tag{B.8}$$

where $\delta_1(s_t)$ and $\delta_2(s_t)$ with $s_t = 1, 2$ are shape parameters. The parameters $\delta_1(s_t)$ and $\delta_2(s_t)$ influence both asymmetry and tail-heaviness of $\varepsilon_t(s_t)$. (ii) The regime-dependent conditional mean of y_t is $\mu_t(s_t) + \exp[\lambda_t(s_t)]\{\Psi^{(0)}\{\exp[\delta_1(s_t)]\} - \Psi^{(0)}\{\exp[\delta_2(s_t)]\}\}$. (iii) The regime-dependent conditional volatility of y_t is $\exp[\lambda_t(s_t)]\{\Psi^{(1)}\{\exp[\delta_1(s_t)]\} + \Psi^{(1)}\{\exp[\delta_2(s_t)]\}\}^{1/2}$, where $\Psi^{(0)}(x)$ and $\Psi^{(1)}(x)$ are digamma and trigamma functions, respectively. (iv) The regime-dependent log conditional density of y_t is

$$\begin{aligned}\ln f(y_t|y_1, \ldots, y_{t-1}, s_t) = {} & \exp[\delta_1(s_t)]\varepsilon_t(s_t) - \lambda_t(s_t) \\ & - \ln\Gamma\{\exp[\delta_1(s_t)]\} \\ & - \ln\Gamma\{\exp[\delta_2(s_t)]\} + \ln\Gamma\{\exp[\delta_1(s_t)] + \exp[\delta_2(s_t)]\} \\ & - \{\exp[\delta_1(s_t)] + \exp[\delta_2(s_t)]\}\ln\{1 + \exp[\varepsilon_t(s_t)]\}\end{aligned} \tag{B.9}$$

(v) The regime-dependent score function is given by:

$$\begin{aligned}u_{\lambda,t}(s_t) = {} & \{\exp[\delta_1(s_t)] + \exp[\delta_2(s_t)]\}\frac{\varepsilon_t(s_t)\exp[\varepsilon_t(s_t)]}{\exp[\varepsilon_t(s_t)] + 1} \\ & - \exp[\delta_1(s_t)]\varepsilon_t(s_t) - 1\end{aligned} \tag{B.10}$$

MS NIG-EGARCH

(i) The error term has the normal-inverse Gaussian distribution:

$$\varepsilon_t(s_t) \sim \mathrm{NIG}\{0, 1, \exp[\delta_1(s_t)], \exp[\delta_1(s_t)]\tanh[\delta_2(s_t)]\} \tag{B.11}$$

where $\delta_1(s_t)$ and $\delta_2(s_t)$ with $s_t = 1, 2$ are shape parameters. The parameters $\delta_1(s_t)$ and $\delta_2(s_t)$ influence tail-heaviness and asymmetry of $\varepsilon_t(s_t)$ (the asymmetry parameter is given by $\exp[\delta_1(s_t)]\tanh[\delta_2(s_t)]$).

(ii) The regime-dependent conditional mean of y_t is:

$$\mu_t(s_t) + \frac{\exp[\lambda_t(s_t)]\tanh[\delta_2(s_t)]}{\{1 - \tanh^2[\delta_2(s_t)]\}^{1/2}} \tag{B.12}$$

(iii) The regime-dependent conditional volatility of y_t is:

$$\left\{\frac{\exp[2\lambda_t(s_t)-\delta_1(s_t)]}{\left\{1-\tanh^2[\delta_2(s_t)]\right\}^{3/2}}\right\}^{1/2} \tag{B.13}$$

(iv) The regime-dependent log conditional density of y_t is:

$$\begin{aligned}\ln f(y_t|y_1,\ldots,y_{t-1},s_t) &= \delta_1(s_t)-\lambda_t(s_t)\\ &-\ln(\pi)+\exp[\delta_1(s_t)]\{1-\tanh^2[\delta_2(s_t)]\}^{1/2}\\ &+\exp[\delta_1(s_t)]\tanh[\delta_2(s_t)]\varepsilon_t(s_t)\\ &+\ln K^{(1)}\left\{\exp[\delta_1(s_t)]\sqrt{1+\varepsilon_t^2(s_t)}\right\}-\frac{1}{2}\ln[1+\varepsilon_t^2(s_t)]\end{aligned} \tag{B.14}$$

where $K^{(1)}(x)$ is the modified Bessel function of the second kind of order 1. (v) The regime-dependent score function is given by:

$$\begin{aligned}u_{\lambda,t}(s_t) &= -1-\exp[\delta_1(s_t)]\tanh[\delta_2(s_t)]\varepsilon_t(s_t)\\ &+\frac{\varepsilon_t^2(s_t)}{1+\varepsilon_t^2(s_t)}\\ &+\frac{\exp[\delta_1(s_t)]\varepsilon_t^2(s_t)}{\sqrt{1+\varepsilon_t^2(s_t)}}\\ &\times\frac{K^{(0)}\left\{\exp[\delta_1(s_t)]\sqrt{1+\varepsilon_t^2(s_t)}\right\}}{+K^{(2)}\left\{\exp[\delta_1(s_t)]\sqrt{1+\varepsilon_t^2(s_t)}\right\}}\\ &\qquad 2K^{(1)}\left\{\exp[\delta_1(s_t)]\sqrt{1+\varepsilon_t^2(s_t)}\right\}\end{aligned} \tag{B.15}$$

where $K^{(j)}(x)$ is the modified Bessel function of the second kind of order j.